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Finite Sample Spaces cont'd: dispositions, Maxwell-Boltzmann statistics etc.

placement / order matters

Ω_D = dispositions of k balls into n boxes

Ω_{MB} = k balls into n boxes (functions)

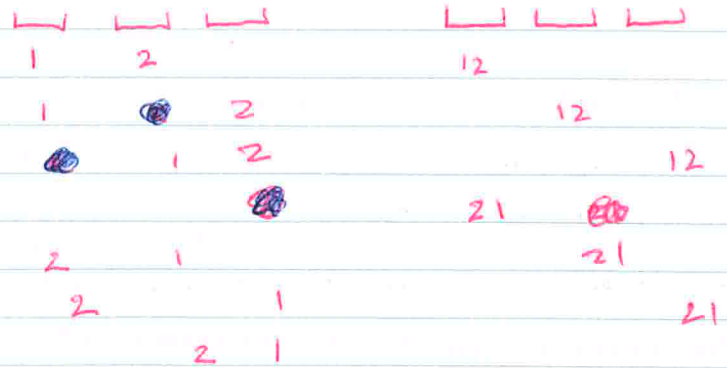
$$|\Omega_{MB}| = n^k \; ; \; P_{MB}(\omega) = \frac{1}{n^k}$$

$$|\Omega_D| = \langle n \rangle_k = n(n+1)\cdots(n+k-1)$$

$$P_D(\omega) = \frac{1}{\langle n \rangle_k}$$

ex. 2 balls into 3 boxes

$\langle n \rangle_k$ dispositions



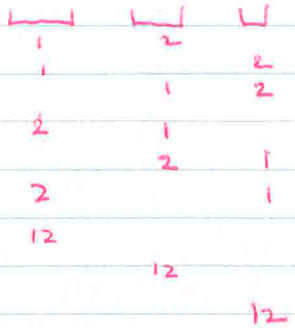
$k=2$, $n=3$

$\langle 3 \rangle_2 = 3\cdot(3+1) = 12$

maxwell-boltzman statistics

ex 2 balls into 3 boxes

n^k, $n=3$, $k=2$ $3^2=9$ ✓



Ex. 3 balls into 4 boxes

dispositions

$\theta_1 = 0 \quad \theta_2 = 1 \quad \theta_3 = 0 \quad \theta_4 = 2$

6 dispositions

1	23
1	32
2	13
2	31
3	12
3	21

$6 = 3!$

Ex. 3 balls into 4 boxes

$\theta_n = 0 \quad 1 \quad 0 \quad 2$

1	23
2	13
3	12

3

\# ways of disposing balls into boxes is always the permutations irrespective of occupation #'s.

$\theta_1 = 1 \quad \theta_2 = 1 \quad \theta_3 = 1 \quad \theta_4 = 0$

$\theta_4 = 3$

\# of dispositions of k balls into n boxes w/ given occupation nos $\theta_1, \theta_2, \dots, \theta_n$ $(\theta_1 + \theta_2 + \dots + \theta_n)$ equals $k!$, irrespective of the θ_i's

Maxwell–Boltzman.

The # of functions from a set B of k elems. to a set U of n elems. with occ. #'s $\theta_1, \theta_2, \theta_n$

$$(\theta_1 + \theta_2 + \theta_3 \dots + \theta_n) = \frac{k!}{\theta_1! \, \theta_2! \dots \theta_n!}$$

$$= \binom{k}{\theta_1, \theta_2, \dots, \theta_n}$$

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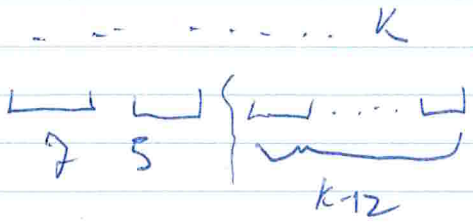
K balls into n boxes

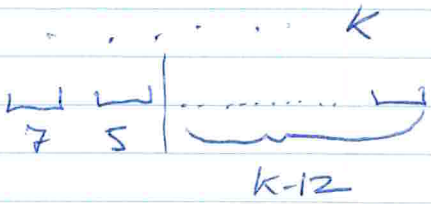
Ex. Disp



$$\frac{K!}{\langle n \rangle_K}$$

mistake on this problem

MB A = event that box 1 contain 7 balls, box 2 5



$$\frac{\binom{K}{7,5,K-12}}{n^K}$$

Ex. Balls in

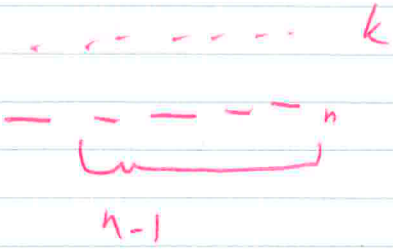
C event that box 1 is empty in Ω_0

$$\frac{\langle n-1 \rangle_k}{\langle n \rangle_k} = \frac{n(n+1)\cdots(n+k-1)}{n(n+1)\cdots(n+k-1)}$$

$$\therefore P(C) = \frac{(n-1)}{(n+k-1)}$$

B = event that box 1 is empty

$P_{MB}(B) =$



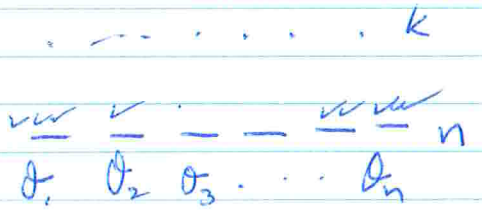
$$\frac{(n-1)^k}{n^k} = P_{MB}(B$$

$$\frac{n-1}{n} = \sqrt[k]{P_{MB}(B}$$

$$\left(1 - \frac{1}{n}\right)^k = \text{[illegible]}$$

Bose-Einstein Stat.

Ω_{BE}



as many checks as balls in boxes

Ex. Boxes are variables call them
$x_1, x_2, \ldots, x_n$

Ω_B = all monomials of total deg. k made with the variables $x_1, \ldots, x_n$

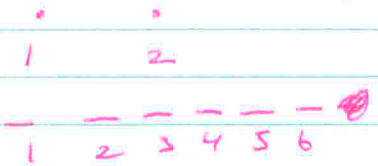
e.g. $x_1^3, x_2^5 \quad x_4^2$ $\qquad 10 = k$
$\qquad 4 = n$

$$|\Omega_{BE}| = ? = \frac{\langle n \rangle_k}{k!}$$

Prob of placing k balls into n boxes w/ an given occ. is

$$\frac{1}{\langle n \rangle_k / k!}$$

Ex. Toss 2 dice, A = event that 7 shows



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Finite Sample Spaces: Bose-Einstein and Fermi-Dirac Statistics

Bose-Einstein

sampling w/ replacement

k indis balls

n boxes

multiset: set where elements can be repeated.

$$\{\underbrace{a, b, c, a, b, a, a}_{k \text{ elements}}\}$$

not a set, a multiset

boxes are elements, balls are multiplicity of those elements.

Warning: can not take union of multi-sets

Ω_{BE} has $\left\langle \binom{n}{k} \right\rangle$ sample pts

multinomial coeff's

$$\left\langle \binom{n}{k} \right\rangle = \frac{\langle n \rangle_k}{k!} = \frac{n(n+1)(n+2)\cdots(n+k-1)}{k!}$$

$$\frac{1}{(1-x)^n} = \sum_{i=0}^{\infty} \left\langle \binom{n}{i} \right\rangle x^i$$

$$\left\langle \binom{n}{k} \right\rangle = (-1)^k \binom{-n}{k}$$

Fermi-Dirac

sampling w/out replacement

k indis balls

n boxes

and all $\theta_i = \{0, 1\}$

Ω_{FD} has $\binom{n}{k}$ sample pts

i.e. $\binom{n}{k} = \binom{n}{k, n-k}$

$$\frac{n!}{k!(n-k)!} = \frac{(n)_k}{k!}$$

placements of distinguishable balls

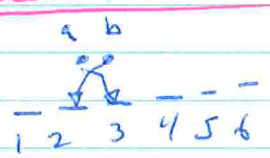
divide by $k!$ since there are $k!$ permutations of these balls in boxes.

Model #1 - picking a subset (Kelvin a

#2 - boxes fruitflies, test one and dies (sampling w/out replacem

$$(1+x)^n = \sum_{i=0}^{n} \binom{n}{i} x^i$$

Back To Dice



MB

A = event that they add to 7

$$P_{MB}(A) = \begin{matrix} (1,6) \\ (2,5) \\ (3,4) \end{matrix} \begin{matrix} \geq \\ \geq \\ \geq \end{matrix} \; 6/36 = \frac{1}{6}$$

$$P_{BE}(A) = \frac{3}{\langle {n \atop r} \rangle} = \frac{3}{\langle {6 \atop 2} \rangle} = \frac{3}{21} = \frac{1}{7}$$

Random Variables – Integer Random Variables on a sample space Ω

ex. Bernoulli Sample Space Ω_B $\quad w = (HTHHT\ldots)$

W = waiting time for the first head (H)

event $\rightarrow P(W = n) = q^{n-1}p$

function $\rightarrow W(w)$ = certain val. no 1, 2, ...

> A random variable x is a function def^n on a sample space w/prop. that taking integ. vals and $(x=n)$ is an event

If x is an integer random variable on a sample space Ω, then the events $(x=n)$ are disjoint for diff n's & furthermore, the $\bigcup_n (x=n) = \Omega \Rightarrow \sum_n P(X=n) = P(\Omega) = 1$

$X(w) = n$

$P(x=n) = P_n$ the sequence P_n is the prob. distribution of the integer random variable x.

$$\boxed{\sum_n p_n = 1 \quad , p_n \geq 0}$$

$$P(W=n) = p_n = q^{n-1} p \qquad n > 0 \quad , P(W=n) = 0 \text{ if } n \leq 0$$

$$\boxed{\sum_{n>0} q^{n-1} p = 1} = p \sum_{n>0} q^{n-1} = p = \frac{1}{1-q} = \frac{p}{p} = 1$$

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<u>Random Variables</u> (Cont'd) X is an integer random variable when $(X=n)$ is an event for every integer n.

The $\boxed{\text{prob. distrib.}}$ of X is the sequence

$p_n = P(X=n)$; $\sum_n p_n = 1$

Two R.V.'s X and Y are indep. when the events $(X=n)$ and $(Y=k)$ are independent for all integers n and k, i.e. when:

$$P((X=n)\cap(Y=k)) = P(X=n)\,P(Y=k)$$

$\boxed{\text{notation}}$ $\rightarrow P = (X=n, Y=k)$

Ex. Bernoulli, x_i and x_j are indep. for $i \neq j$

$$x_i = \begin{cases} 1 & \text{if } i\text{th toss is a success} \\ 0 & \text{failure} \end{cases}$$

$W_1 = T_1$, T_n = gap bet $(n-1)$st and nth success

$W_k = T_1 + T_2 + \cdots T_k$. The R.V.'s T_1 and T are indep

Proof: $P((T_1=n)\cap(T_2=k)) = q^{n-1}p\,q^{k-1}p$; $P(T_1=n) = q^{n-1}p$; $P(T_2=k) = q^{k-1}p$.

X, Y = Random Variables $P((X=n)\cap(Y=k)) = r_{nk} = \boxed{\text{joint dist of the R.V.s } X \text{ and}}$

$P(X=n) = p_n$, $P(Y=k) = q_k$. We have:

$$P(X=n) = \sum_k r_{nk} = \sum_k P((X=n)\cap(Y=k)) = P\Big(\bigcup_k ((X=n)\cap(Y=k))\Big)$$

$$= P\Big((X=n)\cap\underbrace{\Big(\bigcup_k (Y=k)\Big)}_{\Omega}\Big) = P(X=n), \text{ as desired}$$

Similarly, $q_k = \sum_n \mu_{nk}$

If X and Y were indep. then $\mu_{nk} = p_n \cdot q_k$.

Ex

MB

ex. Ans. Joint dist of θ_1 and θ_2

$$P(\theta_1 = i, \theta_2 = j) = \binom{k}{i, j, k-i-j} \frac{(n-2)^{k-i-j}}{n^k}$$

$$P(\theta_1 = i) = \binom{k}{i} \frac{(n-1)^{k-i}}{n^k}$$

$$\sum_{j=0}^{k} P(\theta_1 = i, \theta_2 = j) = P(\theta_1 = i)$$

~~Exp 1:2:th~~

$$\sum_j \binom{k}{i, j, k-i-j} \frac{n-2}{n^k} = \binom{k}{i} \frac{(n-1)^{k-i}}{n^k}$$

$$\binom{k}{i, j, k-i-j} = \binom{k}{i}\binom{k-i}{j}$$

$$\sum_j \binom{k}{i}\binom{k-i}{j} \frac{(n-2)^{k-i-j}}{n^k} = \binom{k}{i} = \sum_j \binom{k-i}{j} \frac{(n-i)^{k-i-j}}{n^k}$$

$$= \binom{k}{i} n^{k} (n-2+1)^{k-i}$$

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Ex.

Ω_{BE} Joint dist. of θ_1 and θ_2

$$P(\theta_1 = i, \theta_2 = j) = \frac{\left\langle {n-2 \atop k-i-j} \right\rangle}{\left\langle {n \atop k} \right\rangle}; \quad P(\theta_1 = i) =$$

$$\Delta = \frac{\binom{n-1}{k-i}}{\left\langle {n \atop k} \right\rangle}$$

$$\sum_j \frac{\left\langle {n-2 \atop k-i-j} \right\rangle}{\left\langle {n \atop k} \right\rangle} = \frac{\left\langle {n-1 \atop k-1} \right\rangle}{\left\langle {n \atop k} \right\rangle}$$

Ω_D again θ_1 and θ_2

$$P(\theta_1 = i, \theta_2 = j) = \binom{k}{i, j, k-i-j} i!\, j! \frac{\langle n-2 \rangle_{k-i-j}}{\langle n \rangle_k}$$

or

$$= \binom{k}{i+j} (i+j)! \frac{\langle n-2 \rangle_{k-i-j}}{\langle n \rangle_k}$$

$$P(\theta_1 = i) = \binom{k}{i} i! \frac{\langle n-1 \rangle_{k-i}}{\langle n \rangle_k} = \sum_j \binom{k}{i, j, k-i-j} i!\, j! \langle n-2 \rangle_{k-i-j}$$

An interesting fact about falling and rising factorials

$$(m+n)_{i+j} = \sum_{i+j=r} \binom{i+j}{i} (m)_i (n)_j = \sum_{i+j=r} \binom{r}{i} (m)_i (n)_j$$

$$(m+n)^r = \sum_{i+j=r} \binom{r}{i} (m^i n^j) \Rightarrow \text{Binomial Thm.}$$

$$\underbrace{\cdots\cdots}_{m}\ \underbrace{\cdots\cdots}_{n} \quad r$$

~~Expectation~~

Ω_{MB} join dist of $\theta_1, \theta_2 \ldots \theta_n$

$$P(\theta_1 = i_1, \theta_2 = i_2, \ldots, \theta_n = i_n) = \binom{k}{i_1, i_2, \ldots, i_n} \frac{1}{n^k}$$

Ω_B $P(\theta_1 = i_1, \ldots, \theta_n = i_n) = \dfrac{k!}{\langle n \rangle_k} = \dfrac{1}{\left\langle {n \atop k} \right\rangle}$

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Expectation of Random Variables

also called the mean

$\rightarrow X$ = integer R.V. $\Rightarrow p_n = P(X=n)$

$$E(X) = \langle X \rangle = \sum_n n p_n = \sum_n n P(X=n)$$

Fundamental Fact:

$$E(X+Y) = E(X) + E(Y)$$

because they are disjoint

$$(X+Y=n) = \bigcup_{i+j=n} (X=i) \cap (Y=j)$$

Proof $E(X+Y) = \sum_n n P(X+Y=n)$

$$= \sum_n n \sum_{i+j=n} P((X=i) \cap (Y=j)) =$$

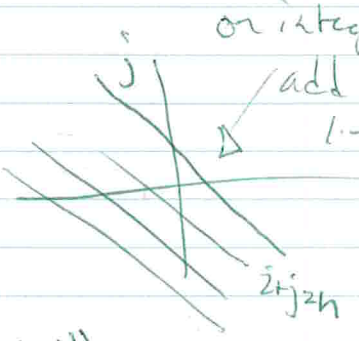
$$= \sum_n \sum_{i+j=n} (i+j) P((X=i) \cap (Y=j))$$

$$= \sum_i \sum_j (i+j) P((X=i) \cap (Y=j)) =$$

$$= \sum_i \sum_j i P((X=i) \cap (Y=j)) + \sum_j \sum_i j P((X=i) \cap (Y=j))$$

$$= \sum_i i \sum_j P((X=i) \cap (Y=j)) + \sum_j j \sum_i P((X=i) \cap (Y=j))$$

$$= \sum_i i P(X=i) + \sum_j j P(Y=j) = E(X) + E(Y), \text{ as desired}$$



on integer coords add all points on line

then have sum over all integer pts on plane

Ex $E(\theta_1)$ in Ω_{MB}

Hardway: $P(\theta_1 = i) = \binom{k}{i} \frac{(n-1)^{k-i}}{n^k}$

$$\Rightarrow E(\theta_1) = \sum_{i=0}^{k} i P(\theta_1 = i) = \sum_{i=0}^{k} i \binom{k}{i} \frac{(n-1)^{k-i}}{n^k}$$

θ_1 and θ_2 are identically distributed

Indetically Distributed Random Variables –

have the same expectation: $E(\theta_1) = E(\theta_2)$

$\therefore E(\theta_3) = E(\theta_1)$ etc

(e.g Bernoulli trials)

$E(\theta_1 + \theta_2 + \dots + \theta_n) = E(\theta_1) + E(\theta_2) + \dots E(\theta_n)$ by basic fact

$X = \theta_1 + \theta_2 + \dots + \theta_n \qquad P(X=k) = 1 \quad P(X=i) = 0$ if $i \neq k$

$E(X) = k \cdot P(X=1) = k$

$E(\theta_1) + \dots + E(\theta_n) = k$

even Ω_{BE} where

$n E(\theta_1) = k = E(\theta_1) = \frac{k}{n} = \sum_{i=0}^{n} i \binom{k}{i} \frac{(n-1)^{k-i}}{n^k} \qquad E(\theta_1) = \frac{k}{n}$

$\cancel{P(\theta_1 = i)}$

Bernoulli

$E(X_n) = 1 \cdot P(X_n = 1) + 0 P(X_n = 0) = p$

$S_n = X_1 + X_2 + \dots + X_n$ = no success in first n trials

$E(S_n) = E(X_1) + E(X_2) + \dots + E(X_n) = np = \sum_{i=0}^{n} \binom{n}{i} \cdot i p^i q^{n-i}$

$E(W_1) = \sum_{i \geq 0} i P(W_1 = i) = \sum_{i \geq 0} i \cdot p q^{i-1} = ? = p \cdot \frac{1}{(1-q)^2} = \frac{p}{p^2} = \frac{1}{p}$

hot way

$\sum_{i \geq 0} q^{i-1} = 1 + q + q^2 + \dots = \frac{1}{1-q}$

$\Rightarrow \sum_{i \geq 0} q^i = 1 + q + q^2 + \dots = \frac{1}{1-q} \xrightarrow{\frac{\partial}{\partial q}} \sum_{i \geq 0} i q^{i-1} = \frac{\partial}{\partial q} \frac{1}{1-q} = \frac{1}{(1-q)^2}$

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W_k = waiting time for k-th success

T_n = gap between (n-1)st and n-th success

$W_k = T_1 + T_2 + \cdots + T_k$

Crucial Remark: T_n and are identically distr

$E(W_k) = E(T_1) + E(T_2) + \cdots + E(T_k) = \frac{k}{p}$

$P(T_n = i) = q^{i-1} p$

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The inclusion-exclusion principle

$X =$ integer r.v. w/ prob. distrib. $p_n = P(X=n)$

$$E(X) = \sum_n n p_n \qquad E(X+Y) = E(X) + E(Y)$$

Indicator Random Variable (simplest type there is)

$$I_A \begin{cases} 1 & \text{if } A \text{ happens} \\ 0 & \text{if } A \text{ don't happen.} \end{cases}$$

$$I_A(w) = \begin{cases} 1 & \text{if } w \in A \\ 0 & \text{if } w \notin A \end{cases}$$

$$E(I_A) = \sum_n n P(I_A = n) = 1 \cdot P(I_A = 1) + 0 P(I_A = 0) = P(A)$$

① $I_{A \cap B} = I_A \cdot I_B$

② $I_{A \cup B} = I_A + I_B - I_A I_B = 1 - (1 - I_A)(1 - I_B)$

③ $I_{A^c} = 1 - I_A$

②+③ $I_{A \cup B} = 1 - I_{A^c} \cdot I_{B^c}$

follows from $(A \cup B)^c = A^c \cap B^c \implies$ for $A_1, A_2, \ldots, A_n \to (A_1 \cup A_2 \cup \ldots A_n)^c = A_1^c \cap A_2^c \cap A_3^c \cdots \cap A_n^c$

$$1 - I_{A_1 \cup A_2 \cup \cdots A_n} = (1 - I_{A_1})(1 - I_{A_2}) \cdots (1 - A_n)$$

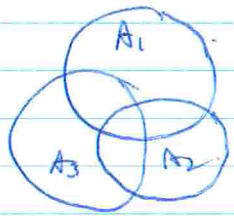
$$I_{A_1 \cup A_2 \cup A_3 \cup \cdots A_n} = 1 - (1 - I_{A_1}) \cdot (1 - I_{A_2}) \cdots (1 - I_{A_n})$$

$$= \sum_{i=1}^{n} I_{A_i} - \sum_{i<j} I_{A_i} \cdot I_{A_j} + \sum_{i<j<k} I_{A_i} \cdot I_{A_j} I_{A_k} - \cdots +$$

taking expectations

$$P(A_1 \cup A_2 \cup \cdots \cup A_n) = \sum_{i=1}^{n} P(A_i) - \sum_{i<j} P(A_i \cap A_j) + \sum_{i<j<k} P(A_i \cap A_j \cap A_k) - \cdots + \cdots + P(A_1 \cap A_2 \cdots$$

The Inclusion-Exclusion Principle



$$P(A_1 \cup A_2 \cup A_3) = P(A_1) + P(A_2) + P(A_3)$$
$$- P(A_1 \cap A_2) - P(A_1 \cap A_3) - P(A_2 \cap A_3)$$
$$+ P(A_1 \cap A_2 \cap A_3)$$

Ex. Two decks of cards the place them down on the table (cards #1-2000) into two decks.
? Prob. in turning over cards that two will simul. show same #.

$1 \quad 2 \quad \cdots \quad n$

$p(1), p(2) \cdots \quad p(n)$

(aka "permutation of n)

equivalent to

Problem: Prob that a random permutation of a set of n elements has at least 1 fixpoint.

A_i = event that ~~permutation~~ $p(i) = i$ (this a point)

$P(A_i) = \frac{(n-1)!}{n!} =$

$P(A_i \cap A_j) = \frac{(n-2)!}{(n)!} = \frac{1}{n(n-1)}$

$$? = P(A_1 \cup A_2 \cup \cdots \cup A_n) = \sum_i P(A_i) \cdots$$

$$= 1 - \binom{n}{2}\frac{1}{n(n-1)} + \binom{n}{3}\frac{1}{n(n-1)(n-2)} - \binom{n}{4}\frac{1}{(n)_4} + \cdots$$

$$= 1 - \frac{n(n-1)}{2!}\cdot\frac{1}{n(n-1)} + \frac{n(n-1)(n-2)}{3}\cdot\frac{1}{n(n-1)(n-2)}$$

$$= 1 - \frac{1}{2!} + \frac{1}{3!} - \frac{1}{4!} + \frac{1}{5!} \cdots \pm \frac{1}{n!}$$

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Recall $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!}$

$x=-1$ $\quad \frac{1}{e} = 1 - 1 + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} + \cdots$

$\boxed{1 - 1/e} = 1 - 1/2! + 1/3! - \frac{1}{4!} + \frac{1}{5!} \simeq P(A_1 \cup \cdots \cup A_n)$

For $n > 4$ good to 3 decimals

$\therefore$ for all practical purposes is constant

ex. MB $\quad P((\theta_1 > 0) \cap \cdots \cap (\theta_n > 0))$

$((\theta_1 > 0) \cap \cdots \cap (\theta_n > 0))^c = \underbrace{(\theta_1 = 0)}_{A_1} \cup \underbrace{(\theta_2 = 0)}_{A_2} \cup (\theta_3 = 0) \cup \cdots \cup \underbrace{(\theta_n =}_{A_n}$

$\sum_{i=0}^{n} P(\theta_i = 0) - \sum_{i<j} P((\theta_i = 0) \cap (\theta_j = 0)) + \cdots$

$P(\theta_1 = 0) = \frac{(n-1)^k}{n^k} = (1 - \frac{1}{n}$

$P((\theta_1 = 0) \cap (\theta_2 = 0)) = \frac{(n-2)^k}{n^k}$

$= \left(\binom{n}{1}(n-1)^k - \binom{n}{2}(n-2)^k + \binom{n}{3}(n-3)^k - \cdots \right) / n^k$

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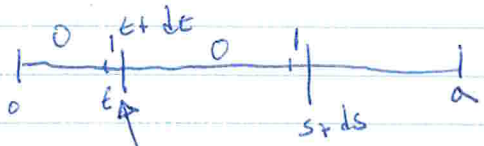
Joint densities of continuous Random Variables

X, Y = continuous R.V.

$$\int_{s=c}^{d}\int_{t=a}^{b} \text{dens}(X=t, Y=s)\,dt\,ds = P((a<X\le b)\cap(c<Y\le d))$$

$$\text{Dens}(X(1)=t, X(2)=s)\,\overbrace{dt\,ds}^{\text{infinitesimal prob.}}$$



$$= \binom{n}{0,1,0,1,n-a}\cdot\frac{dt}{a}\cdot\frac{ds}{a}\cdot\frac{(a-s)^{n-2}}{a^{n-2}}$$

infinitesimal small box allows you to look at this with constraint that only 1 ball falls into infinitesimally small box

indistinguishable marbles = order statistics

The algebra of random variables

X = continuous R.V. w/ density $\text{dens}(X=t) = f(t)$

$2X$ has what density? Well, cumulative probability distrib. of $2X$ is $P(2X\le t)$

but $(2X\le t) = (X\le \frac{t}{2}) = P(X\le \frac{t}{2})$

$$F(t) = \int_{-\infty}^{t} f(s)\,ds = P(X\le t$$

$$\text{dens}(2X=t) = \frac{d}{dt}F\left(\frac{t}{2}\right)$$

$$= F'\left(\frac{t}{2}\right)\cdot\frac{1}{2} = \underline{\underline{f\left(\frac{t}{2}\right)\frac{1}{2}}}$$

$$\int_{-\infty}^{\infty}\frac{1}{2}f\left(\frac{t}{2}\right)dt = 1$$

ex $X \geq 0$ continuous

$$\text{dens}(X=t) = f(t) = \begin{cases} 0, & \text{if } t<0 \\ \dots \end{cases}$$

$$\text{dens}(X^2=s) = ?, \quad P(X^2 \leq s)$$

cumulative prob distribution

$$(X^2=s) = (X=\pm\sqrt{s})$$

$$F(t) = \int_{-\infty}^{t} f(u)\,du$$

$$\frac{d}{ds} P(X^2 \leq s) = \text{dens}(X^2=s)$$

$$= \frac{d}{ds} F(\sqrt{s}) = F'(\sqrt{s}) \frac{1}{2\sqrt{s}}$$

$$= f(\sqrt{s}) \frac{1}{2\sqrt{s}}$$

$$= \int f(\sqrt{s}) \frac{1}{2\sqrt{s}}\,ds = \int_0^{\infty} f(u)\,du = 1$$

$u = \sqrt{s} \quad du = \frac{1}{\ }$

(2) 18.313, 3/10/98

Application to Expectation

$$E(g(x)) = \int_{-\infty}^{\infty} g(t)\,\text{density}(x=t)\,dt$$

these always give the same answer.

$$\text{dens}(g(x)=s) = h(s)$$

$$E(g(x)) = \int_{-\infty}^{\infty} t\,h(t)\,dt$$

$$E(Y) = \int_{-\infty}^{\infty} t\,\text{dens}(Y=t)\,dt$$

$$\text{dens}(x=t) = f(t)$$

x^n for $x \geq 0$ $\text{dens}(x^n=t) = ?$

$$P(X^n \leq t) = P(X \leq \sqrt[n]{t})$$
$$= F(\sqrt[n]{t})$$

$$\text{dens}(X^n=t) = \frac{d}{dt} F(\sqrt[n]{t})$$
$$= F'(\sqrt[n]{t}) = \frac{1}{n} t^{1/n - 1} = \frac{1}{n} t^{\frac{1-n}{n}}$$

$$\text{dens}(x^n=t) = \frac{1}{n} t^{\frac{1-n}{n}} f(\sqrt[n]{t})$$

$$E(X^n) = \int_0^{\infty} t\,\text{dens}(X^n=t)\,dt = \int_0^{\infty} t \cdot \frac{1}{n} t^{\frac{1-n}{n}} \cdot f(\sqrt[n]{t})\,dt$$
$$= \int_0^{\infty} u^n f(u)\,du = \int_0^{\infty} t^n\,\text{dens}(x=t)\,dt$$